

DAS THERMOMETRY AND REFLECTION SEISMOLOGY: ESTIMATING BASAL PROPERTIES AT THE SOUTH POLE

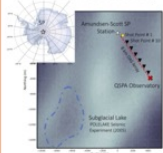
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INTRODUCTION

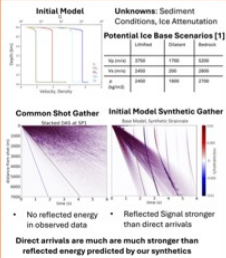
South Pole DAS Array



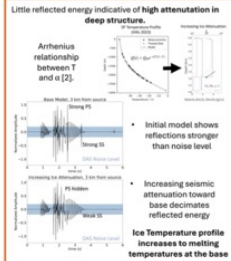
Characterization sedimentary basins and subglacial layers through direct measurements is burdensome but pertinent, as they give insight into the history of the thermal regime and provide invaluable constraints for ice sheet dynamics. Geophysical methods allow indirect measurements of the ice sheet basal conditions through which temperature and a stability characterization can be inferred. In January 2023, we converted the existing South Pole power cable into an 8 km Distributed Acoustic Sensing (DAS) array to perform an active source survey. Through reflection and attenuation modeling of our data, we find the presence of a weak PS phase to be consistent with an ice column temperature profile which increases to melting conditions at the base; further, the ice column overlies a likely thick dilatant sediment basin. These results will allow for a better constraint on relative contributions of basal sliding and subglacial deformation to the estimated basal slip at the South Pole.

Keywords: distributed acoustic sensing, glacial seismology, thermometry

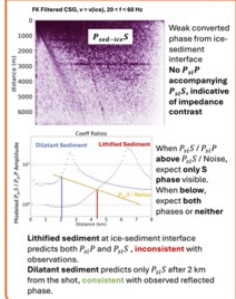
DATA AND SYNTHETICS



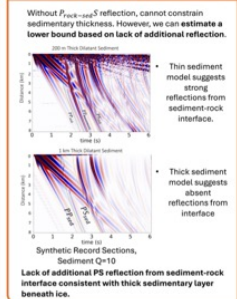
ENGLACIAL TEMPERATURE



SEDIMENTARY COMPOSITION

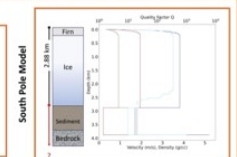


SEDIMENTARY THICKNESS



CONCLUSION

- ### Key Takeaways
- Ice temperature is increasing to melting temperatures at the base, consistent with radar inferred profile.
 - Weak reflected phases are consistent with high attenuation near the ice base
 - Ice sheet overlies a dilatant (wet, porous) sediment layer.
 - The relative strength of PS phases without a PP present is most consistent with a dilatant sediment
 - The sediment layer is likely thick. This is consistent with MT and Refraction estimates.
 - Lack of strong reflections from the sediment base with increasing thickness of the layer



REFERENCES

- [1] Peters, L.E., Aranderkhan, S., Holland, E.W., Horgan, H.J., Björnsdóttir, D.D., Völgt, D.E., 2008. "Seismic detection of a subglacial lake near the South Pole, Antarctica." Geophysical Research Letters.
- [2] Hsiao, B.H. et al., "Geophysics and Thermodynamics at South Pole Lake Indicate Stability and a Regionally Thawed Bed." Geophysical Research Letters.

FUTURE RESEARCH

- Sedimentary Thickness:** Current work involves using receiver functions at QSPA Broadband Station.
- Continuous Passive Monitoring:** Soon, a year of passive seismic data from the South Pole DAS array will be returned. We are particularly interested in the near surface seasonal fluctuations and teleseismic recordings.

CONTACT INFORMATION

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